

1. U attributes

`u.attrs`

Same ordered group `J_sel`

Y, I_Y
pos(I_Y, J_sel)
→ `lhs_contract`

Z, I_Z
pos(I_Z, J_sel)
→ `rhs_contract`

2. Establish I_U

```
f_Y = ordered_complement(
    ι(I_Y), u.attrs.lhs_contract)

f_Z = ordered_complement(
    ι(I_Z), u.attrs.rhs_contract)
```

```
I_U = gather(I_Y, f_Y)
      ++ gather(I_Z, f_Z)
```

Extend axis mappings to U

Preserved groups

`K` `J_rem` `I_sel` `H`

3. Final attributes

`out.attrs`

Same ordered group `G = I_sel ++ K`

X, I_X
pos(I_X, G)
→ `lhs_contract`

U, I_U
pos(I_U, G)
→ `rhs_contract`

Constructed contraction attributes

Attribute	u.attrs	out.attrs
lhs_contract	pos(I_Y, J_sel)	pos(I_X, G)
rhs_contract	pos(I_Z, J_sel)	pos(I_U, G)

pos(I, G) returns q with gather(I, q) = G.
ι(I) = range(|I|); ++ concatenates sequences.